NOVA IMS INFORMATION MANAGEMENT SCHOOL

OCTOBER 2026

Everything & Then Some

Will They Return, and How Much Will They Spend?

Group Project
ETAI 2026/27



I. “Everything & Then Some”

Everything & Then Some, Ltd. is a diversified wholesale company supplying businesses and specialist retailers across a broad portfolio of products, including tools and hardware, lighting, automotive care, household goods and other specialist product lines. Its customers show very different purchasing patterns, making repeat-purchase behaviour difficult to anticipate.

The company wants to use its historical sales data to support more targeted commercial actions. Your team has been asked to develop a deployment-ready solution capable of answering two questions: **will this customer make another purchase within the next 90 days after the current transaction, and how much will they spend in that period?**

Together, the two answers let the commercial team act on each customer: those unlikely to return can be targeted with incentives to bring them back, while those expected to spend the most can be prioritized for retention and dedicated account management. Your tool should therefore work for any new purchase, giving the team both predictions at the moment it matters.

II. PROJECT OBJECTIVES

1. **Data Exploration and Cleaning**: explore the supplied transaction-level data, identify its inconsistencies and fix the data errors (e.g., invalid values, duplicates).
2. **Classification Pipeline Optimization**: predict whether a customer will make *another purchase within 90 days of the current purchase*. Define a consistent model-assessment strategy and optimize the full pipeline: pipeline-dependent preprocessing (e.g., imputation, encoding, scaling, outlier treatment), feature engineering, feature selection, algorithms and hyperparameters.
3. **Kaggle Competition**: generate predictions for every observation in test.csv and submit them using the provided sample_submission.csv structure.
4. **Regression Pipeline Optimization**: for the customers your classifier predicts will return, predict *how much they will spend in the 90 days after the current purchase*. You decide how to approach the problem and how to combine it with the classification, ensuring both predictions are coherent.
5. **Deployment:** build a pipeline for each task that runs from the terminal, and a client interface that, for a new purchase, returns whether the customer will buy again and how much they will spend.
6. **Explainability and Open-Ended Section**: explain the predictions of both models with the explainability methods covered in class (e.g., SHAP, LIME). Beyond that, be as creative as you want: this section should complete your project and help it stand out. Examples include a dashboard for the commercial team, a complete user interface for the tool, or further analyses (e.g., prediction errors, ablation study, stability over time, calibration, or uncertainty of the spend predictions).

III. PROJECT DATA

You have access to three data files and one sample submission file.

- ***train.csv*** contains 6,534 historical purchase occasions, the binary target *repeat_purchase_90d* and, for customers who purchase again, the regression target *spend_next_90d*.
- **test.csv** contains 986 later purchase occasions with the same predictors but without the targets.
- **clientes.csv** is an optional customer-level table that can be joined through customer_id .
- **sample_submission.csv** provides the required Kaggle submission structure and a simple baseline prediction.

Each row in train/test represents a customer purchase occasion. The available data contains the following attributes and attribute groups

Variable	Description
ID	Unique modelling-row and Kaggle submission identifier
customer_id	Anonymized customer identifier; join key for clientes.csv .
repeat_purchase_90d	Classification target: 1 if the customer purchases again within 90 days; 0 otherwise. Train only.
spend_next_90d	Regression target: total amount (EUR) the customer pays in their later purchases within 90 days after the current one. Train only; **never use it as a feature for classification.**
purchase_date	Anonymized purchase date. Temporal order and intervals are preserved.
Current monetary values	Current transaction totals, tax, shipping, discounts and cost.
Current purchase details	Current detail count and number of invoices in the purchase occasion.
Current categoricals	Payment, salesperson, zone, transport, warehouse and document series.
Historical timing	Recency, tenure, prior purchase count and historical purchase gaps.

Historical value	Prior spend and prior purchase-value summary statistics.
Rolling frequency	Purchase counts over the previous 30, 90, 180 and 365 days.
Rolling spend	Customer spend over the previous 30, 90, 180 and 365 days.
random_noise	Additional numerical field; assess whether it contains useful information.

IV. DELIVERABLES

Upon the project's deadline, you will be required to submit:

1. A zip file named **GroupXX_ETAI_2627.zip** (XX = your group number) with all the code you used throughout the project to:
   a. Decide on your final solutions for both tasks.
   b. Obtain your final results. Code that helped you make decisions but does not directly contribute to the final solution (e.g., exploration notebooks) should be included only if relevant.
2. A **GitHub repository** (link) with the final pipelines, a *README.md* explaining how to run it on Mac and Windows, a *config.yaml* and a *requirements* file.
3. Your **final Kaggle submission**, matching the output of your pipeline.

The pipeline

The pipeline must run entirely from the terminal, without changing any code. You can structure it as you prefer, as long as it contains:

- **Training for each task**, from raw data to a saved model, so that predicting never retrains.
- **Batch prediction** for test.csv, writing the Kaggle submission (ID, repeat_purchase_90d) and the regression predictions (ID, repeat_purchase_90d, spend_next_90d) for the rows predicted to return.
- **The client interface**: for one new purchase, the probability and decision of a repeat purchase and the expected spend in the next 90 days.
- A **config.yaml** with every choice that can change without editing code (e.g., paths, seed, models, hyperparameters).

V. EVALUATION

Your final project will be graded out of **20v**, according to the following criteria:

Criteria	Percentage (%)	Max Grade (out of 20)
Pipeline Structure & Code Quality	15	3
Data Exploration & Cleaning	15	3
Classification Pipeline Optimization	25	5
Regression Pipeline Optimization	20	4
Explainability & Open Ended Section	15	3
Deployment	10	2

- Students can submit all deliverables with a **maximum delay of 3 days**, incurring a penalty of **1 point per day**. Beyond these three days, submissions will not be accepted.
- Deliveries made before the deadline will receive a **bonus of 0.15 points per day** of delivery in advance (up to a **maximum of 1 point**).

Your grade will reflect our assessment of the quality of your work in terms of quality of writing, clarity, conciseness, correctness and efficiency:

- **Pipeline Structure and Code Quality (3v):** organization of the repository, readability, documentation and reproducibility.
- **Data Exploration & Cleaning (3v):** meaningful insights, and clear steps and rationale behind the data cleaning (errors, inconsistencies, invalid values, duplicates).
- **Classification Pipeline Optimization (5v):**
  - Kaggle performance: 1v
  - Model assessment strategy (minimum of 5 algorithms): 1v
  - Pipeline-dependent preprocessing and feature engineering: 1v
  - Feature selection strategy and implementation: 0.75v
  - Hyperparameter tuning: 0.75v
  - Choice of evaluation metric(s) and interpretation of results: 0.5v

- **Regression Pipeline Optimization (4v):**
  - Target exploration and modelling approach: 0.75v
  - Model assessment strategy and a minimum of 4 algorithms: 0.75v
  - Pipeline-dependent preprocessing, feature engineering, feature selection and tuning: 1.5v
  - Coherence between classification and regression: 0.5v
  - Performance on the test set (RMSE, MAE and R^2 on customers who actually returned: 0.5v
- **Explainability & Open-Ended Section (3v):**
  - Explainability of both models, globally and for individual customers: 1.5v;
  - Formulation and difficulty of any additional objectives: 0.5v;
  - Correctness of implementation: 0.5v;
  - Discussion of results and their business meaning: 0.5v.
- **Deployment (2v):**
  - The pipeline runs from the terminal as described (1v);
  - The client interface works for new and returning customers (1v).

VI. FINAL NOTES

1. Features must be computable at the time of the purchase, and each test row must be predicted using only that row, the training data and clientes.csv (never other test rows).
2. If you use automated packages (e.g., ydata-profiling), we will only consider observations that have a visible impact on the project (e.g., a key insight that is acted upon).
3. You may use scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, SHAP, LIME, imbalanced-learn, TensorFlow, Keras and PyTorch. Using Lazy Predict or similar AutoML (e.g., Feature Tools, TSFresh) packages is explicitly forbidden.
4. We won't consider any steps or results that are not in your zip file or repository.
5. Refrain from providing theoretical explanations of topics covered in class.
6. Source information from peer-reviewed journals (avoid Medium, TowardsDataScience and similar sources). Do not cite ChatGPT or any other LLM.
7. We will clone and run your repository. Please ensure that every command runs without interruption. Repositories that do not fulfil this condition will be severely penalized.
8. The code will pass through a process of plagiarism and AI generation checking.
9. You must submit to the Kaggle competition to get points for that component.
10. When determining the grade for your work, there will be a comparative component between it and the work presented by your peers.

Defense: each group has a **40-minute defense**. At the start, the professors randomly decide the order in which members present. Each member explains what the group did and why, and the professors may ask any member questions about any part of the code. Not attending will result in a 0 in the project grade.
The grade of each element of a group can differ depending on personal performance, **without limits**.

Friendly Reminders:

1. If something is good enough to be mentioned, it is also good enough to know. DO NOT include techniques/algorithms/steps you cannot explain: we may (and probably will) ask about them in the discussion.
2. Finished is better than perfect.

Project Delivery: 22nd December, 18h00